

AIR-range: Arranging optical systems to present mid-AIR images with continuous luminance on and above a tabletop

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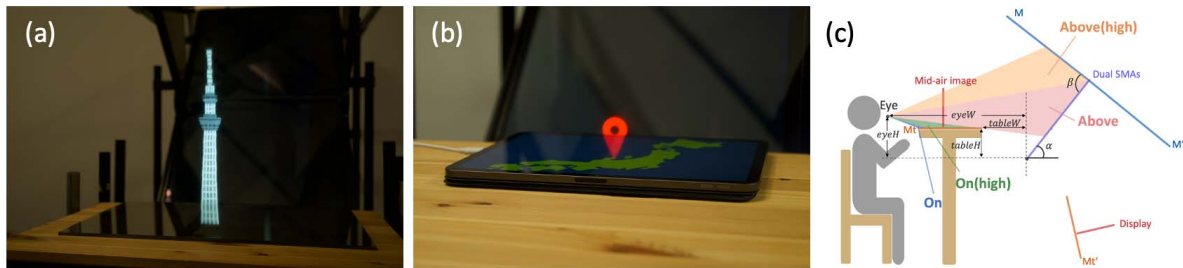


Figure 1: Applications and system overview of AIR-range. (a) Tall mid-air image of TOKYO SKYTREE. (b) Mid-air image of map pin on iPad. (c) The proposed arrangement of optical systems.

ABSTRACT

We propose “AIR-range”- a system that seamlessly connects mid-air images from the surface of a table to mid-air space. This system can display tall mid-air images in the three-dimensional (3D) space beyond the screen. AIR-range is implemented using a symmetrical mirror structure that displays a large image by integrating multiple imaging paths. The mirror arrangement in previous research had a problem in that the luminance was discontinuous. In this study, we theorize the relationship between the parameters of optical elements and the appearance of mid-air images and optimize an optical system to minimize the difference in luminance between image paths.

Index Terms: Human-centered computing— Human computer interaction (HCI)— Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

Tables are useful as surfaces for providing an augmented-reality environment. A table with mid-air images acts as a virtual surface where users can interact with the virtual contents as they would with physical objects. Dual slit-mirror arrays (dual SMAs) [2], composed of two layers of SMA, form a real image at a position symmetrical to the light source, and image does not have any distortions depending on the viewing angle. Dihedral corner reflector array [6] has a different structure but behaves similarly. The dual SMAs have a recommended viewing angle, and in the case of ASKA3D¹ used in this system, the recommended angle is $45^\circ \pm 20^\circ$.

Some systems using the dual SMAs were proposed to expand the range of mid-air image presentation. Optical systems including MARIO [4] and Hashimoto and Murofushi's optics [1] can display

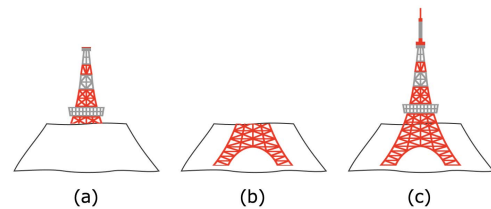


Figure 2: Comparison of a tall mid-air image on the table from the user's perspective. (a) Mid-air image on the table surface. (b) Mid-air image above the table surface. (c) Mid-air image on and above the table surface.

mid-air images above a tabletop (Fig. 2 (a)). In contrast, systems such as EnchanTable [7] and PortOn [5] were proposed for displaying vertically standing mid-air images on a glossy surface (Fig. 2 (b)). Kajita et al. [3] proposed a method that displays images on and above a tabletop, as show in Fig. 2 (c), but it can cause a part of the mid-air image to vanish. Yokose et al. [8] proposed a display using a symmetrical mirror structure that enables mid-air images to be displayed on and above a tabletop without interruption (Fig. 2 (c)). However, in this method, a part of the mid-air image may decrease its luminance because the viewing angle of the dual SMAs deviates from the recommended angle.

In this study, we proposed an optical system that can minimize the discontinuity of luminance by changing the position and angle of mirrors and present a clear mid-air image over a wide area, from the table surface to mid-air space. By designing the system so that the user does not feel the constraint caused by the optical system, we can give the user a surprise that the image is projected in the air, encouraging interaction.

2 SYSTEM DESIGN AND EXPERIMENT

The optical system of the AIR-range is illustrated in Fig. 1 (c). The system consists of a table, a display, the dual SMAs, a half-mirror Mt placed on the table, mirror Mt' arranged symmetrically with respect to Mt and the dual SMAs, and symmetric mirrors M and M'.

We categorized areas by the position of the imaging paths as

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¹<https://aska3d.com/ja/>

Table 1: Constraints and results of parameters from the optimization.

	Constraints	Results
α	$0^\circ - 90^\circ$	51.5°
β	$0^\circ - (90 + \frac{\alpha}{2})^\circ$	89.4°
$tableW$	$0 - 263\text{mm}$	220mm
$tableH$	$0 - 146\text{mm}$	146mm
Viewing angle at Above(high) area	$31.7^\circ - 61.5^\circ$	$47.6^\circ - 61.1^\circ$
Viewing angle at Above area	$25.0^\circ - 62.0^\circ$	$31.3^\circ - 48.6^\circ$
Viewing angle at On(high) area	$45.0^\circ - 56.1^\circ$	$45.7^\circ - 55.0^\circ$
Viewing angle at On area	Not considered	Not considered

follows:

Above (high) area area imaged by rays passing in the order of Display-M'-dualSMAs-M-Eye (orange area in Fig. 1 (c)).

Above area area imaged by rays passing in the order of Display-dualSMAs-Eye (pink area in Fig. 1 (c)).

On (high) area area imaged by rays passing in the order of Display-Mt'-dualSMAs-Mt-Eye (green area in Fig. 1 (c)).

On area area imaged by rays passing in the order of Display-Mt'-M'-dualSMAs-M-Mt-Eye (blue area in Fig. 1 (c)).

Ideally, the luminance should be the same for the entire mid-air image; however in reality, it differs with the viewing angle and area. In this study, we first set a constraint to align the luminance ranges of the areas, and then optimized the mirror arrangement to minimize the differences in luminance between the areas.

The angle at which the mid-air image is observed was determined by the following parameters: the angle of the dual SMAs to the horizontal plane (α), the viewing point (($eyeW, eyeH$)), the angle of M to the dual SMAs (β), and the position of Mt (($tableW, tableH$)).

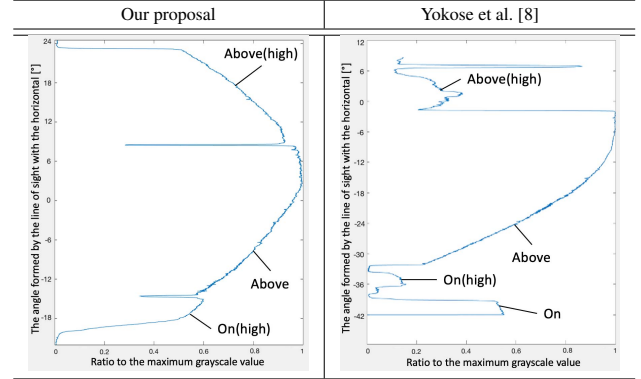
The ranges of these parameters were determined as the constraints (Table 1). These constraints ensured that all the areas were observed with the same luminance as when Above area is observed at the recommended angle. We decided not to consider On area because it was invisible when optimized. Instead, On (high) area extended down to On area and presented a mid-air image on the table. The viewpoint position was set to ($eyeW, eyeH$) = (670mm, 203mm). Here, we used the least-squares technique of data fitting to fit a trigonometric function to the relationship between luminance and viewing angle. Because the viewing angle is determined by the parameters, we can calculate the mirror arrangement for luminance. The parameters that minimized the total difference in luminance were calculated by solving the constrained optimization for a nonlinear multivariable function. The values of parameters calculated by the optimization technique are listed in Table 1.

Table 2 summarizes the results of comparing the luminance of the optical system with mirrors arranged by these parameters and those by the conventional method. The horizontal axis of the graph represents the ratio of the grayscale value at each viewing angle to the maximum grayscale value, and the vertical axis corresponds to Above (high) area to On (high) area from top to bottom. The graphs show that the differences in luminance within each area and between areas were dwindling, and the overall uniformity of luminance was improved. The results revealed that the areas were smoothly connected, and the differences in luminance at the boundaries of the areas were nearly zero.

3 APPLICATION

The AIR-range can produce mid-air images across the On (high), Above, and Above (high) areas. Fig. 1 (a) illustrates a mid-air image

Table 2: Proposed method vs. previous method. It shows that the luminance of proposed method is more continuous.



of TOKYO SKYTREE, which has been difficult to display with previous systems. The tower appears seamlessly from the table surface to mid-air. In addition, installing a tabletop display under On (high) area enables the superimposition of mid-air images onto the image of the display. Figure 1 (b) illustrates a map of Japan displayed on an iPad and the mid-air image of a map pin superimposed on it. Because the AIR-range guarantees luminance of the mid-air image presented in the On (high) area, it enables the projection of both the image on the tablet tabletop display and the mid-air image.

4 CONCLUSION

We proposed a method to optimize the arrangement of optical systems and showed improvement in the luminance continuity of the mid-air image on and above a tabletop.

ACKNOWLEDGMENTS

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